

Rain Zimin Yang

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EDUCATION

Columbia University

Sept 2026 - now

Ph.D in Computer Science.

Advised by: Rocco Servedio, Toniann Pitassi, Xi Chen and Josh Alman.

University of British Columbia

Sept 2023 - May 2026

BSc Combined Honours in Computer Science and Mathematics with Distinction.

- GPA: 92.8%
- Undergraduate courses: Theory of Computation (CPSC 421), Randomized Algorithms (CPSC 436R), Quantum Computation (CPSC 436Q), Group Theory (MATH 322), Number Theory (MATH 437/538), Measure Theory (MATH 420/507), Probability (MATH 418/544), Fields and Galois Theory (MATH 422/501)
- Graduate courses: Commutative Algebra (MATH 423/502), Algebraic Topology (MATH 427/527), Algebraic Geometry (MATH 532), Algebraic Number Theory (MATH 538), Communication Complexity (CPSC 536Z, audit)

RESEARCH

Rational Degree Conjecture

Apr 2025 - Apr 2026

UBC Computer Science | Supervisor: [Daochen Wang](#)

- We proved that a natural complexity measure, the rational degree, of total Boolean functions is polynomially related to its Fourier degree, resolving the Rational Degree Conjecture posed in 1994.
- Accepted to the 67th Annual Symposium on Foundations of Computer Science ([FOCS 2026](#)).

Boundary Regularity for Solutions to the Plateau Problem

May - Aug 2024

UBC Mathematics | Supervisor: Ailana Fraser

- Investigated the existence of stable minimal surfaces with finite total curvature and straight-line boundary whose limit is the half-helicoid using tools from complex analysis.
- Developed MATLAB visualizations to illustrate surfaces using Weierstrass-Enneper representation.
- Presented in the [UBC USRA Seminar](#) on June 20, 2024.
- Research supported via NSERC WLIURA.

ACADEMIC COMPETITIONS

- **Honors in ICPC World Finals 2025** Sept 2025
1st place in Canada, 8th place in North America (Team *Forgetful Functors*)
- **11th place in ICPC NAC 2026** (Team *Diamond Dust*) Mar 2026
- **ICPC Pacific Northwest Regional Contest 2025 - 2026** Nov 2025
Bronze Medalist, Canadian Site Champion (Team *Diamond Dust*)
- **ICPC Pacific Northwest Regional Contest 2024 - 2025** Nov 2024
Silver Medalist, Canadian Site Champion (Team *Forgetful Functors*)
- **William Lowell Putnam Mathematical Competition 2025** Dec 2025
Top 431 out of 4329 Participant (90th percentile of competitors)

TECHNICAL SKILLS

Programming Languages: C++, Java, Python

Mathematical Tools: MATLAB, Mathematica

Other: $\text{\LaTeX}$, Git

AWARDS AND SCHOLARSHIPS

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| • Srivani Family Doctoral Fellowship , <i>Columbia University</i> | 2026 |
| • Dean of Science Scholarship (\$220), <i>UBC Faculty of Science</i> | 2026 |
| • Rick Sample Memorial Award (\$2500), <i>UBC Computer Science</i> | 2026 |
| • D F MacKenzie Scholarship (\$1100), <i>UBC Faculty of Science</i> | 2026 |
| • NSERC Undergraduate Student Research Award , <i>UBC Computer Science</i> | 2024, 2025, 2026 |
| • Stanley M Grant Scholarship in Mathematics (\$4500), <i>UBC Mathematics</i> | 2025 |
| • Dean of Science Scholarship (\$210), <i>UBC Faculty of Science</i> | 2025 |
| • Reginald Palliser-Wilson Scholarship (\$4400), <i>UBC Mathematics</i> | 2024 |
| • Entrance Scholarship in Mathematics (\$10,424.12), <i>UBC Mathematics</i> | 2023 |
| • Outstanding International Student Scholarship (\$13,000), <i>UBC Faculty of Science</i> | 2023 |
| • COMC Entrance Scholarship (\$1000), <i>UBC Mathematics</i> | 2023 |

SERVICES

Conference review: ITCS (2027)

Teaching Assistant

- COMS 4236 Introduction to Computational Complexity (Columbia, Fall 2026)
- CPSC 420 Advanced Algorithm Design and Analysis (UBC, Spring 2026)
- CPSC 421 Introduction to the Theory of Computation (UBC, Fall 2025)
- CPSC 320 Intermediate Algorithm Design and Analysis (UBC, Summer 2025, Fall 2024)